



Blockchain Infrastructure for Intelligent Cyber-Physical-Social Systems

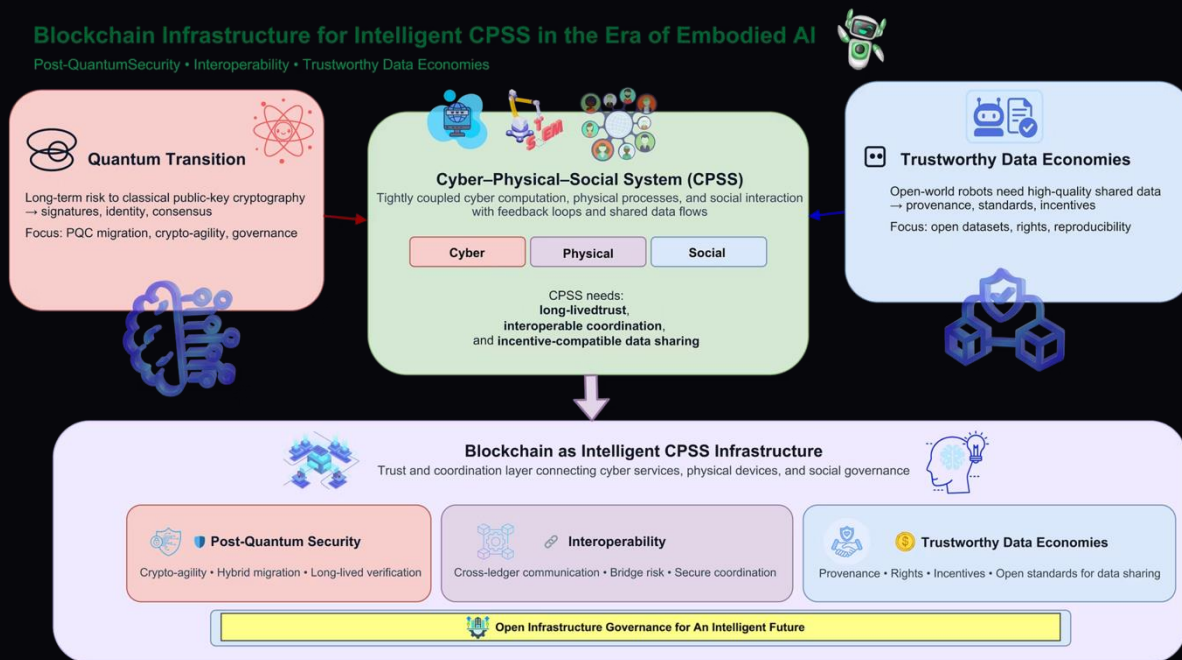
Post-Quantum Security • Interoperability • Trustworthy Data Economies in the Era of Embodied AI

IEEE ICBC 2026 TUTORIAL

IEEE COMMUNICATIONS SOCIETY

QUT Gardens Theatre • Brisbane, Australia • 1–5 June 2026

A Convergence of Three Transformative Transitions



QBE paradigm unifies three urgent infrastructure imperatives: quantum-resistant cryptography, scalable cross-shard coordination, and incentive-compatible data governance — all converging on blockchain as the foundational coordination layer for next-generation Cyber-Physical-Social Systems.

Why Now

The Dual Transition Creating an Infrastructure Imperative



Quantum Threat

Advances recognized by the **2025 Nobel Prize in Physics** and the **Turing Award** now put classical public-key cryptography — signatures, identity, consensus — at credible long-term risk.



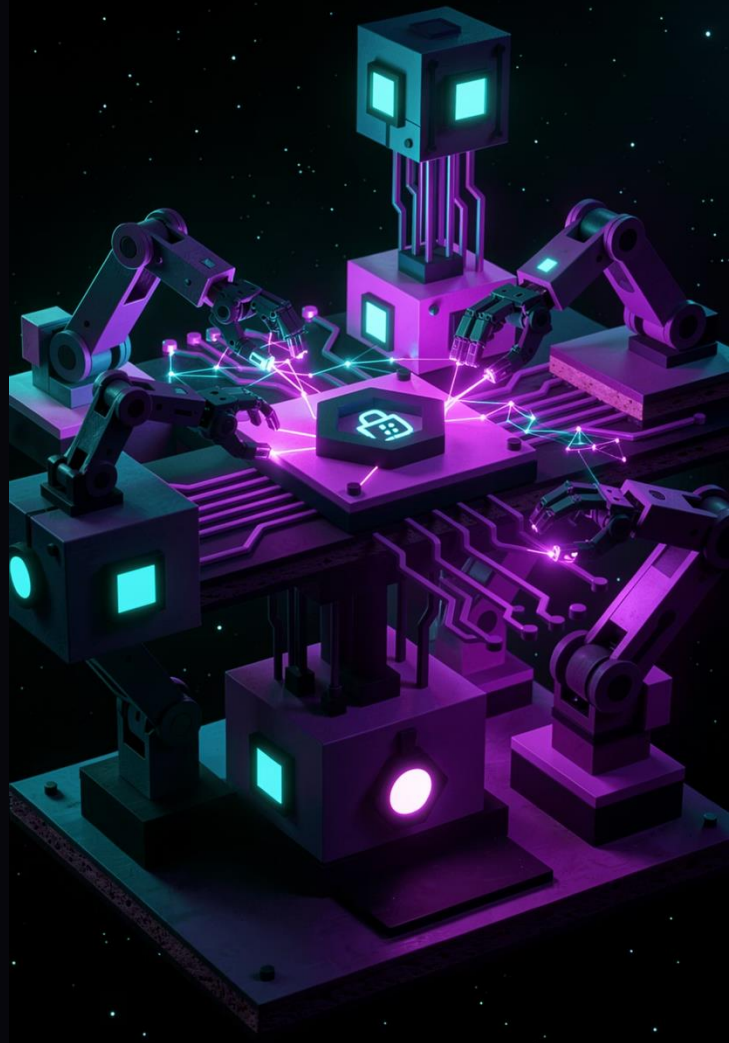
Embodied AI Demand

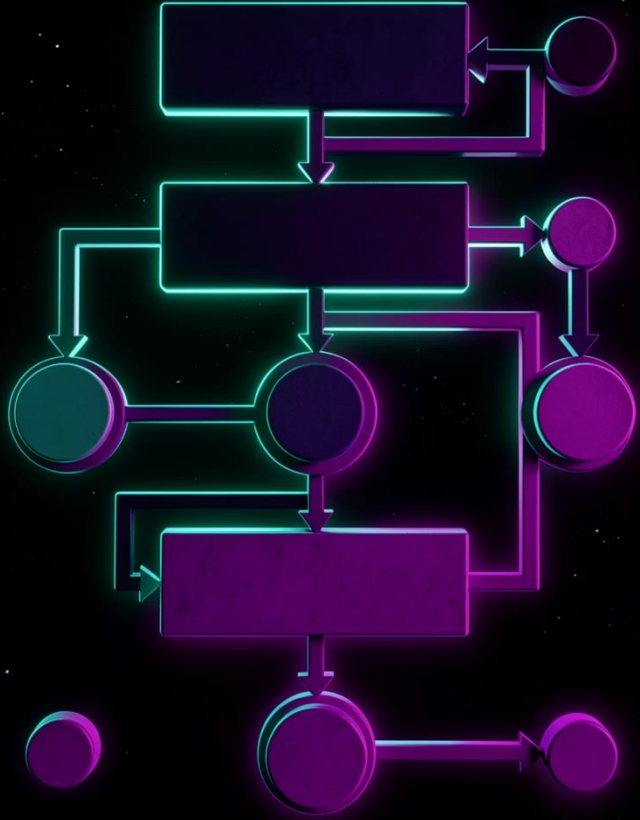
World-model robotic systems operating in open environments generate massive interaction data that requires **trustworthy provenance**, cross-organizational governance, and incentive-compatible sharing.



Blockchain's Role

Blockchain must evolve beyond financial substrate into a unified **CPSS coordination layer** — resisting quantum cryptanalysis while enabling scalable, trustworthy data economies.





Tutorial Structure

Five Integrated Modules + Live Demo



Demo Launch

Open AWS Braket quantum threat baseline



M1 · World Models

Embodied AI and simulated environments



M2 · Hardware Risk

Assess quantum hardware security threats



M3 · BrokerChain

Design scalable cross-shard architecture



M4 · Data Economies

Trustworthy governance and incentive design



M5 · Industry Integration

From theory to production deployment

Each module transitions logically into the next — from empirical quantum security assessment through scalable cross-shard coordination to production-ready governance — forming a cohesive pipeline for CPSS infrastructure design. Duration: **2 hours**, closing with an integrated panel synthesis.

Module 1 · Song Guo

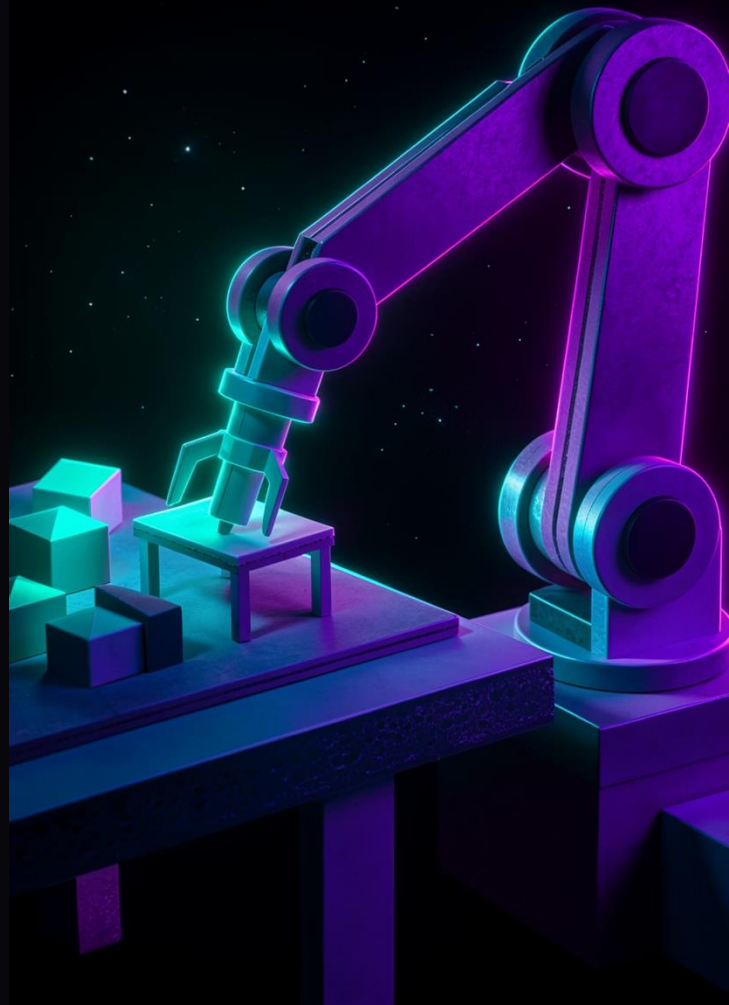
Embodied AI and World Models

Key Focus Areas

- **IRASim**: A fine-grained world model for robot manipulation, enabling simulation of complex interaction trajectories at ICCV 2025
- Edge-distributed robotic systems operating in unpredictable open-world environments
- Massive multimodal interaction data requiring **trustworthy provenance** and incentive-compatible sharing
- Defines the target infrastructure that subsequent modules must secure and scale

Infrastructure Requirements

Embodied AI systems require **long-lived verification**, interoperable coordination across heterogeneous platforms, and incentive-compatible data-sharing mechanisms — all of which depend on crypto-agile blockchain architectures.



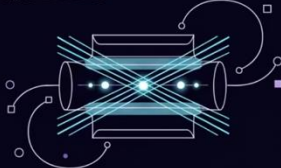
Quantum Hardware: Live on AWS Braket

SUPERCONDUCTING QUBITS

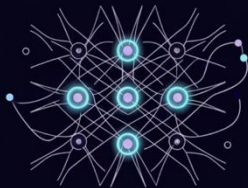


**Fast gate operations.
Limited coherence times.**

TRAPPED-ION QUBITS



**High fidelity quantum gates.
Slower processing.**



**Scalable atomic arrays.
Mid-range coherence.**

NEUTRAL-ATOM QUBITS

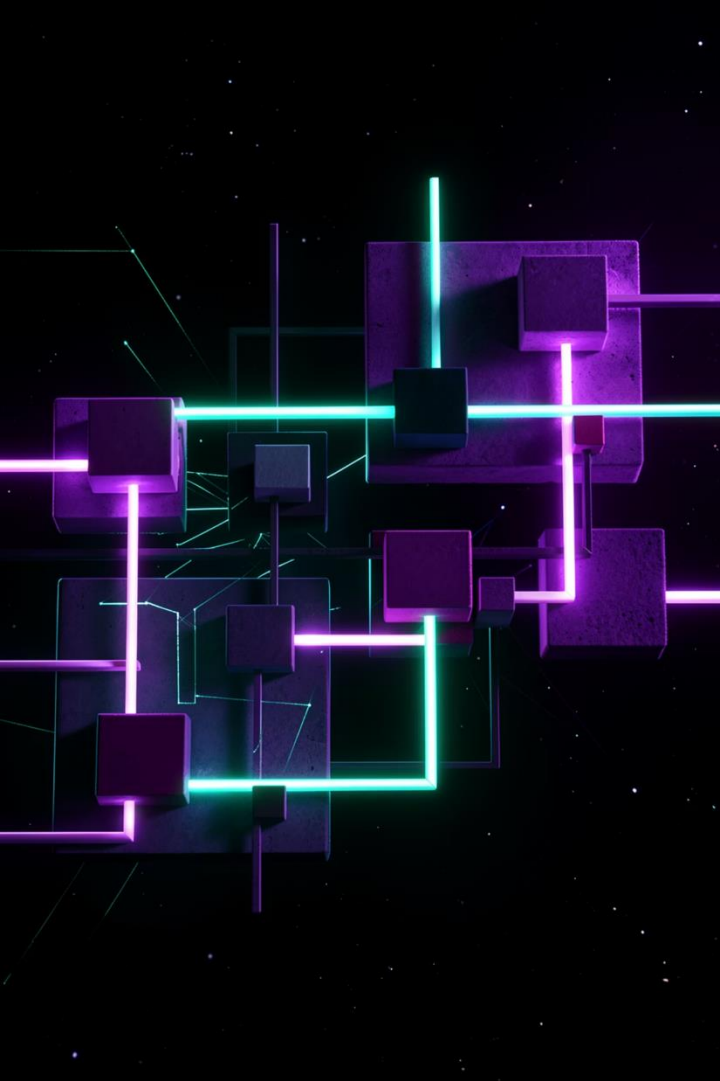


**Scalable atomic arrays.
Mid-range coherence.**

The tutorial opens with an **immersive live demonstration** on Amazon Braket, empirically engaging participants with all three leading quantum hardware modalities.

- Empirical assessment of **ECDSA vulnerability timelines** across hardware classes
- Witnessed signature transition: **ECDSA → post-quantum schemes** (NIST PQC standards)
- Evidence-based migration strategy grounded in real hardware benchmarks
- Hybrid quantum-classical algorithms and cross-domain blockchain security applications

 Keynote delivered at the 2025 Forum on Cloud Quantum Computing Services for Blockchain Security.



Module 3 · Huawei Huang

Scalable Architecture and Interoperability via BrokerChain

BrokerChain Protocol

Cross-shard blockchain protocol for account/balance-based state sharding, published at **IEEE INFOCOM 2022** and extended in *IEEE Transactions on Networking* (2025). Academic Testnet launched June 2025.

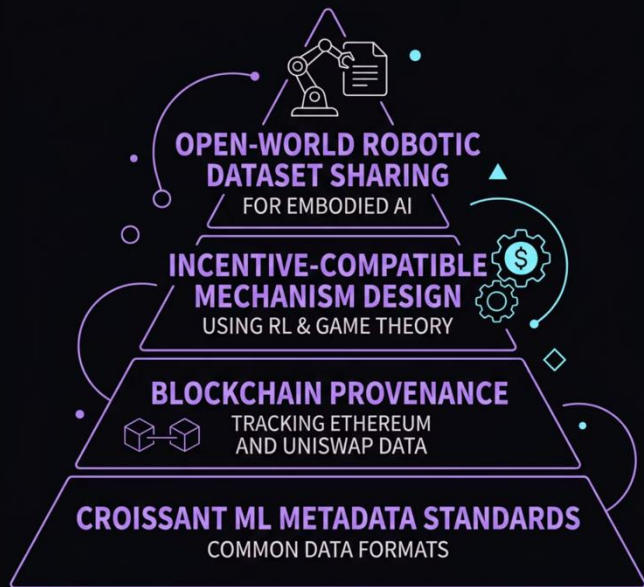
GriDB Scaling

Scales blockchain databases via sharding and off-chain cross-shard mechanisms, published in **VLDB 2023**. Provides the data-layer foundation for high-throughput CPSS deployments.

BlockEmulator Platform

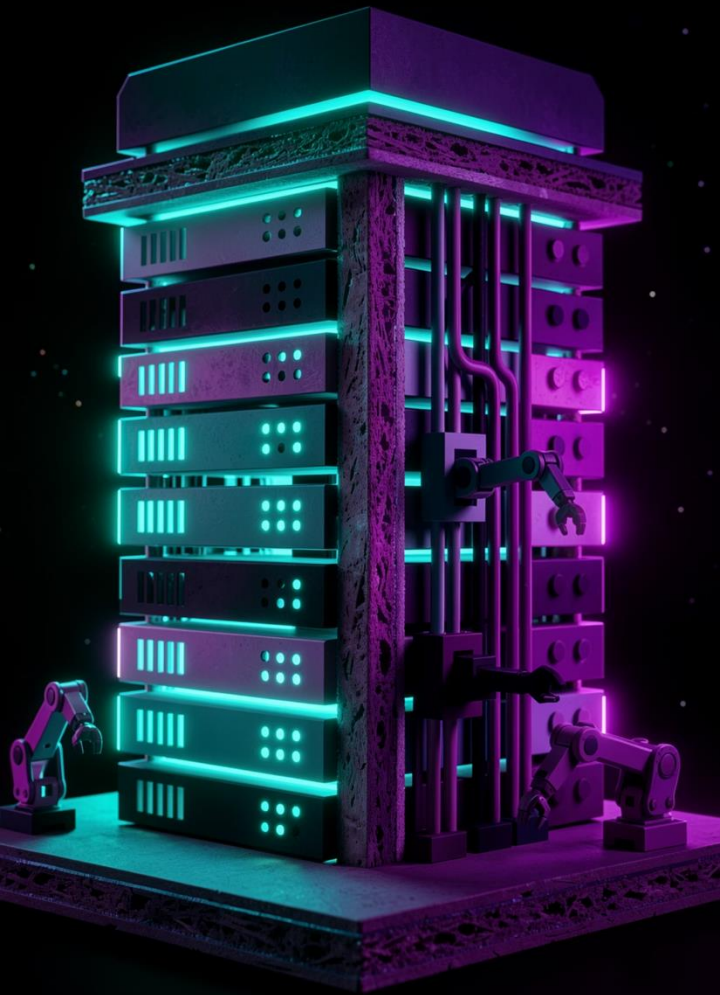
Open-source experimental blockchain platform supporting researchers in **90+ countries**, enabling reproducible cross-shard protocol evaluation and bridge security analysis.

Trustworthy Data Economies



This module implements the governance layer atop BrokerChain's scalable foundation:

- **Croissant** metadata standard (NeurIPS 2024) for ML-ready dataset provenance
- Blockchain provenance for **Ethereum Beacon Chain** rewards and **Uniswap** transaction indices (*Scientific Data*, 2025)
- Proof-of-Stake incentive redesign via reinforcement learning and mechanism design
- Security analysis of cross-ledger interoperability (IEEE S&P 2024 SoK)



Module 5 · Dongping Liu

Industry Ecosystem Integration: From Theory to Production



Multimodal Cloud Deployment

AWS-based infrastructure for deploying embodied AI datasets at scale, bridging academic research with enterprise-grade HPC and AI workloads.



LET & Kuavo Datasets

Real-world robotic learning datasets (LejuRobotics LET Dataset; ICRA Kuavo Data Challenge) operationalized as production-ready CPSS data pipelines.



End-to-End Governance

Completes the pipeline: quantum-resistant security → scalable coordination → trustworthy provenance → production deployment for next-generation CPSS.

Tutorial Speakers

An Interdisciplinary Team Across Blockchain, AI, Quantum, and Economics



Song Guo

Chair Professor, HKUST. IEEE Fellow, ACM Distinguished Member. Editor-in-Chief, *IEEE Transactions on Cloud Computing*. Edward J. McCuskey Award (2024).



Huawei Huang

Professor, Sun Yat-sen University. IEEE Senior Member. Stanford Top 2% Scientists (2021-). Lead developer of BrokerChain and BlockEmulator. 9,000+ citations.



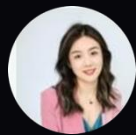
Dongping Liu

CEO, Tenorshare; former AWS Senior BDM. Ph.D. Physics, CAS. 15 years bridging academic research with cloud AI/HPC deployments. Beijing Science & Technology Award (First Prize).



Aoyu Zhang

Senior Applied Scientist, AWS China. Specializes in quantum cloud solutions, hybrid quantum-classical algorithms, and blockchain security applications. Keynote speaker at the 2025 Forum on Cloud Quantum Computing Services for Blockchain Security. Led the AWS Braket integration for quantum hardware benchmarking across superconducting, trapped-ion, and neutral-atom platforms.



Luyao Zhang

Assistant Professor of Economics and Senior Research Scientist at the Digital Innovation Research Center, Duke Kunshan University. Publications in ReStat, IEEE S&P, ACM CCS, NeurIPS, and Nature Scientific Data. NSF-funded researcher and Secretary of the IEEE P3469 Working Group on Blockchain-based Digital Asset Management. Research spans mechanism design, trustworthy data economies, blockchain provenance, and open-science infrastructure for AI datasets.

IEEE ICBC 2026

Register & Join Us at the 8th IEEE International Conference on Blockchain and Cryptocurrency

Conference Details

IEEE ICBC 2026 is the 8th edition of the premier IEEE venue for blockchain and cryptocurrency research, sponsored by the **IEEE Communications Society**.

This tutorial is designed for researchers, engineers, and policy-makers working at the intersection of blockchain, AI, quantum computing, and cybersecurity.

What You Will Take Away

- Open-source experimental frameworks (BlockEmulator, AWS Braket notebooks)
- Actionable roadmaps for quantum-resistant, interoperable CPSS infrastructure
- Hands-on experience with superconducting, trapped-ion, and neutral-atom quantum hardware
- A unified perspective on post-quantum security, scalable sharding, and trustworthy data economies



All participants receive open-source code and experimental frameworks developed by the tutorial team.